

# Danny Rollo

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## RESEARCH INTERESTS

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Deep learning for medical imaging and healthcare; machine learning for urban planning and public transit; computer vision and representation learning for football (soccer); recommender systems and cold-start generalization; imitation learning for robotic manipulation; audio and music signal processing; agentic LLM systems.

## EDUCATION

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### Northeastern University

Boston, MA

*M.S. in Computer Science*

Sept 2024 – April 2026

*B.S. in Computer Science & Mathematics*

Sept 2021 – May 2025

- *Graduate coursework:* Applied Deep Learning, Deep Learning, Machine Learning, Algorithms, Empirical Research Methods, Web Development, Computer Graphics
- *Undergraduate coursework:* AI4Impact, Matrix Methods in Machine Learning & Data Analysis, Macroeconomics, Systems Security, Object-Oriented Design, Statistics & Stochastic Processes

## PUBLICATIONS

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### Peer-Reviewed Conference

1. **Rollo, D.** “Lightweight CNN Encoders for Nucleus Segmentation.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026. **Accepted — poster.**

A controlled encoder ablation showing a 34M-parameter VGG U-Net matches MedT — a transformer purpose-built for small medical datasets — at a  $\sim 24\times$  smaller parameter budget, and outperforms pretrained Swin-T by 4.9 Dice points with  $2.6\times$  less memory and  $1.8\times$  lower inference latency.

### In Preparation

2. **Rollo, D.**, et al. “Two-Tower Recommendation with Cold Start Ablation.” *Manuscript in preparation.*

A 12-experiment ablation on a dual-encoder retrieval model over 1.3M Yelp reviews, with a leave-one-out onboarding protocol that prevents test-set leakage on cold users.

## RESEARCH EXPERIENCE

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### Lightweight CNN Encoders for Nucleus Segmentation

2025–2026

*Sole author · Northeastern University*

PyTorch, Hugging Face timm, PanNuke, MoNuSeg

- Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
- Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.

- Attributed the result to spatial-resolution preservation at skip connections and convolutional inductive bias outweighing Swin’s pretraining advantage in the low-data regime.
- Authored the full manuscript, produced all quantitative and qualitative figures, and wrote the discussion.

## **Flow Matching for Robotic Manipulation**

2025

*Team of 3 · Northeastern University*

PyTorch, MuJoCo, MetaWorld

- Implemented continuous-time flow matching for imitation learning: a neural vector field transports Gaussian noise to expert action distributions via RK2 ODE integration, avoiding the iterative denoising cost of diffusion policies.
- Scaled from a proprioception-only baseline on single-object bin-picking to an RGBD- and language-conditioned policy with 8-step action chunking across 3-object scenes; characterized compounding-error limits on multi-object generalization.

## **Two-Tower Recommendation with Cold Start Ablation**

2025

*Primary contributor & project lead · team of 4*

PyTorch, BPR loss, Yelp Open Dataset

- Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
- Designed a leave-one-out evaluation protocol preventing train/test leakage, applied to the cold-start case where a user’s only signal comes from held-out reviews.
- Built a 4-stage data pipeline (extract, preprocess, split, feature prep) with iterative interaction-density filtering and stratified cold-start holdout; reduced dataset construction from 47s to 8s via vectorization.
- Implemented on-device feature indexing eliminating per-batch CPU–GPU transfer, with per-epoch resampling of geographic negatives and Gumbel-max onboarding selections as data augmentation.

## **RailMax: Amtrak Service Optimizer**

2025

*Open-source modeling study*

InterpretML, GTFS, Census ACS, Folium

- Modeled the US Amtrak network as a directed weighted graph (edges = weekly trips between station pairs) and engineered node-level features including destination-population-weighted reachability and eigenvector centrality.
- Trained an Explainable Boosting Machine ( $R^2 = 0.76$ , RMSE 0.84 log scale) under 5-fold stratified group cross-validation with k-means geographic clustering to prevent spatial leakage.
- Integrated Census ACS, GTFS, NTAD, and IPEDS sources; deployed an interactive what-if frequency simulator identifying the top-20 city pairs most likely to benefit from added service.

## **NJ Transit Reliability Tracker: Independent OTP Measurement**

2026

*Sole author · live public deployment*

Cloudflare Workers, GTFS-Realtime, SQL, TypeScript

- Built and deployed a live public reliability tracker for NJ Transit commuter rail on Cloudflare Workers, continuously ingesting GTFS-Realtime arrivals and recomputing on-time performance at stricter thresholds than the agency’s published 6-minute definition (88% official vs. 80.6% measured at 5 minutes).
- Reported full delay distributions, time-of-day and day-of-week heatmaps, and transfer-connection reliability in place of a single headline rate, with a significance gate suppressing spurious week-over-week movements on low-frequency lines.
- Unified three sources of differing cadence — GTFS static, monthly published performance reports back to 2017, and the realtime feed — with per-figure provenance labelling separating official from

independently measured numbers.

### **Hopewell Road Sketches: Parametric Street-Safety Modeling**

2026

*Sole author · open source*

Python, geopandas, shapely, trimesh, Blender

- Built a parametric geometry pipeline (geopandas, shapely, trimesh) that generates to-scale 2D and 3D before/after street-design comparisons for real intersections from NJDOT road data, OpenStreetMap geometry, and field measurements.
- Computed safety outcomes from first principles — crossing distance, pedestrian time-in-roadway at MUTCD 3.5 ft/s, AASHTO turn speed at tightened corner radii, and parking loss from paint geometry — with every input traced to an authoritative source rather than a geometric default.
- Enforced cross-representation consistency with regression and property-based tests, surfacing geometry defects between the 2D and 3D pipelines that visual inspection had missed.

### **Hopewell Area Crash & Crosswalk Map**

2026

*Sole author · open source, in public use*

Leaflet, Python, OpenStreetMap, GitHub Pages

- Built an interactive crash and pedestrian-infrastructure map for Hopewell Borough, NJ, combining incident records with a personally conducted field survey of 84 crosswalks across 44 locations.
- Designed a repaving priority score integrating crash-harm proximity, survey condition grades, and municipal paving cycles, and derived crossing geometry from OpenStreetMap with a road-centerline fallback for unmapped crossings.

### **Booker**

2025

*Sole author · live demo*

Python, Claude API, MongoDB Atlas Vector Search, MCP

- Built a multi-agent system for artist–venue booking matching: a coordinator agent with bounded iterations routes to specialist agents (artist, venue, booking advisor) via structured, typed tool calls against the Claude API, preventing runaway agent loops by construction.
- Implemented cross-modal semantic retrieval for artist/venue matching over MongoDB Atlas Vector Search using `all-mpnet-base-v2` embeddings (768-dim), with a 3-tier memory system (conversation, working, preferences) for context-aware multi-turn interactions.
- Enforced content safety and domain boundaries with NeMo Guardrails, exposed the system via an MCP server for standardized Claude Desktop tool interop, and instrumented it with OpenTelemetry tracing.

## **PROFESSIONAL EXPERIENCE**

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### **Pitchcut — Founder & CTO**

June 2026 – Present

*Remote*

- Built a computer-vision pipeline (tracking, event/goal detection, player re-identification) for grass-roots football match footage, validated against SoccerTrack v2 ground truth, with a consent-gated data flywheel for continued training-set collection.
- Optimized the CV pipeline’s CUDA inference path (single-GPU-upload tile cropping for 17-tile SAHI-style detection, fp16 autocast) for a  $2.4\times$  throughput gain (245ms  $\rightarrow$  102ms/frame) over the CPU-tile fp32 baseline, auto-detected across CUDA/MPS/CPU and parity-verified before shipping.
- Built submitit-managed SLURM training infrastructure under Explorer’s single-GPU-per-job cap: throttled job arrays with checkpoint/resume, node-throughput profiling that auto-excludes slow nodes (26.8 img/s V100 vs. 3.75 T4, a  $7.1\times$  gap), OAuth-storm-tuned retry/backoff, and MLflow logging that survives SIGKILL-on-timeout.

- Architected the surrounding multi-tenant SaaS platform (Postgres/Supabase, row-level tenant isolation) and an agent-driven back office (Claude Code skills) automating registration, roster, and content drafting with human-in-the-loop approval.

**Teradata — AI/ML Product Management Analytics Intern**

June 2025 – May 2026

*San Diego, CA (Remote)*

- Authored the business case and led the design of a context-aware AI Notebook Assistant for Vantage-Cloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
- Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.

**PlateMate — Co-Founder & Chief Technology Officer**

May 2025 – Present

*Boston, MA*

- Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus cross-modal semantic retrieval over restaurant reviews.
- Built a representation-learning retrieval pipeline with 768-dimensional `all-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.

**LineVision Inc. — Software Engineer Co-op**

Jan 2024 – June 2024

*Boston, MA*

- Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
- Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.

**Brekinridge Capital Advisors — Infrastructure Intern**

July 2023 – Sept 2023

*Boston, MA*

- Cleaned datasets, built Python web scrapers, and trained a clustering model on the resulting data.
- Performed Windows and Linux system administration and security work using PowerShell, Darktrace, and Bit9; built front-ends in React.js and Razor Pages.

**Generate Product Development**

Jan 2023 – May 2025

*Boston, MA*

*Project Lead*

Dec 2024 – May 2025

- Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.

*Technical Lead*

Sept 2023 – Dec 2023

- Led a team of developers building an iOS/Android app on a React Native frontend with a Go/Postgres backend.

*Software Engineer*

Jan 2023 – May 2024

- Backend developer on a team of 8 building ReMo, a literacy tool for educators; designed the database and internal model and implemented the networking API in Go.

**TEACHING EXPERIENCE**

**Khoury College of Computer Sciences — Lead Teaching Assistant**

2022 – 2025

- Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.
- Taught recursion, linked lists, and functional programming (Fundamentals I), then Java object-oriented design, polymorphism, data structures, algorithm design, and time complexity (Fundamentals II).

## HONORS & AWARDS

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### Rookie of the Year, HackBeanpot

February 2023

Built a food-desert tracking web application on a six-person team.

### Students of the Year Runner-Up, Leukemia & Lymphoma Society

March 2021

Raised \$58,000 for blood cancer research in seven weeks during the height of the COVID-19 pandemic.

### Eagle Scout, Scouts BSA

October 2019

Led 30 volunteers to design, fundraise, and construct six portable benches for a local school, delivered on schedule despite pandemic restrictions and supply shortages.

## TECHNICAL SKILLS

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**Languages & Core Libraries:** Python, Go, TypeScript/JavaScript, SQL, pandas, NumPy

**Deep Learning:** PyTorch, Hugging Face timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention, flow matching, diffusion policies, imitation learning

**Machine Learning:** scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering, two-tower retrieval, ablation design, cross-validation methodology

**LLM & Agentic Systems:** Claude and OpenAI APIs, MCP servers, multi-agent orchestration, prompt engineering, RAG, SentenceTransformers, vector search, NeMo Guardrails, OpenTelemetry

**Infrastructure:** Docker, AWS (S3), CUDA, SLURM job arrays, submitit, MLflow, FastAPI, MongoDB Atlas, Git

**Statistical & Research Computing:** Stata (coursework), R (basic), scipy, LaTeX, Unix/Linux, SLURM / HPC clusters

**Data & Visualization:** Folium, Streamlit, GTFS, Census ACS, librosa

## LANGUAGES & ACTIVITIES

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**Languages:** Latin (limited working); English (native).

**Activities:** Generate Product Development (Northeastern student-run product studio); captain, Northeastern FC (BSSL) and intramural soccer; guitarist and songwriter, Northeastern Jazz Combo and Music Farm.

References available upon request.